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PART A - Git & Application Setup

Install Kubectl:

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> $stableVersion = (curl.exe -s https://dl.k8s.io/release/stable.txt).Trim()
PS C:\WINDOWS\system32> $stableVersion
v1.37.0
PS C:\WINDOWS\system32> curl.exe -LO "https://dl.k8s.io/release/$stableVersion/bin/windows/amd64/kubectl.exe"
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 60.51M 100 60.51M 0 0 6.94M 0 00:08 00:08 5.63M
PS C:\WINDOWS\system32> mkdir -Force C:\kubectl
PS C:\WINDOWS\system32> Move-Item -Force .\kubectl.exe C:\kubectl\

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----          03-09-2026    22:36          kubectl

PS C:\WINDOWS\system32> [Environment]::SetEnvironmentVariable("Path", [Environment]::GetEnvironmentVariable("Path", [EnvironmentVariableTarget]::Machine) + ";C:\kubectl", [EnvironmentVariableTarget]::Machine)
PS C:\WINDOWS\system32> $env:Path = [System.Environment]::GetEnvironmentVariable("Path", "Machine") + ";" + [System.Environment]::GetEnvironmentVariable("Path", "User")
PS C:\WINDOWS\system32> kubectl version --client
Client Version: v1.33.1
Kustomize Version: v5.6.0
PS C:\WINDOWS\system32>
```

Install Minikube:

```
PS C:\WINDOWS\system32> choco install minikube
Chocolatey v2.4.2
Installing the following packages:
minikube
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading Minikube 1.38.1... 100%

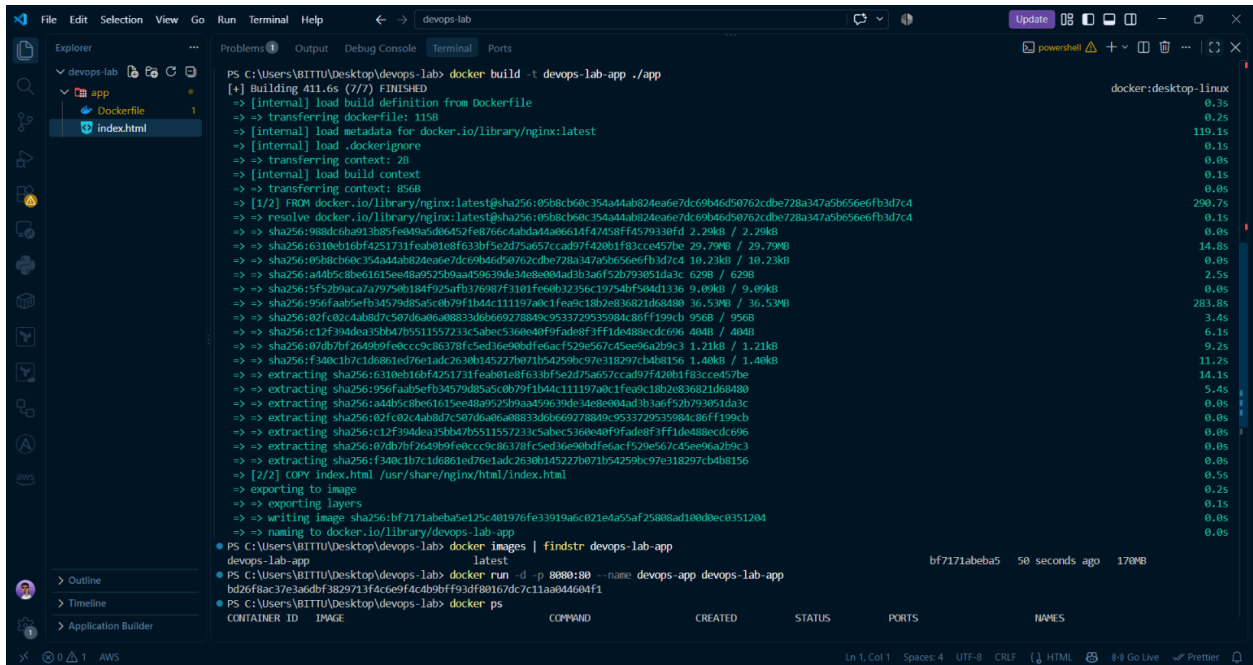
Minikube v1.38.1 [Approved]
Minikube package files install completed. Performing other installation steps.
ShimGen has successfully created a shim for minikube.exe
The install of Minikube was successful.
Deployed to 'C:\ProgramData\chocolatey\lib\Minikube'

Chocolatey installed 1/1 packages.
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\WINDOWS\system32> minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
```

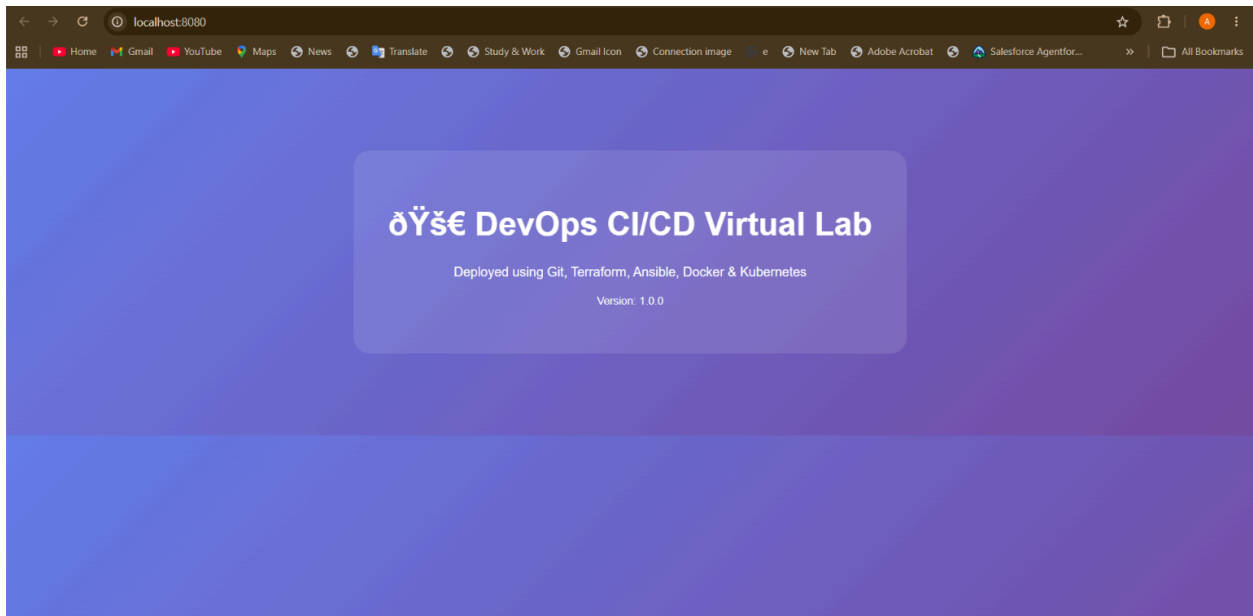
DEMO APP

```
File Edit Selection View Go Run Terminal Help
devops-lab
app > index.html > ...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>DevOps CI/CD Lab</title>
5 <style>
6   body {
7     font-family: Arial, sans-serif;
8     margin: 50px;
9     background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
10    color: white;
11    text-align: center;
12  }
13  .container {
14    background: rgba(255,255,255,0.1);
15    padding: 40px;
16    border-radius: 20px;
17    max-width: 600px;
18    margin: 100px auto;
19  }
20  h1 { font-size: 2.5em; }
21 </style>
22 </head>
23 <body>
24   <div class="container">
25     <h1> DevOps CI/CD Virtual Lab</h1>
26     <p>Deployed using Git, Terraform, Ansible, Docker & Kubernetes</p>
27     <p><small>Version: 1.0.0</small></p>
28   </div>
29 </body>
30 </html>
```

DOCKER IMPLEMENTATION: App turned to Docker Img for containerisation

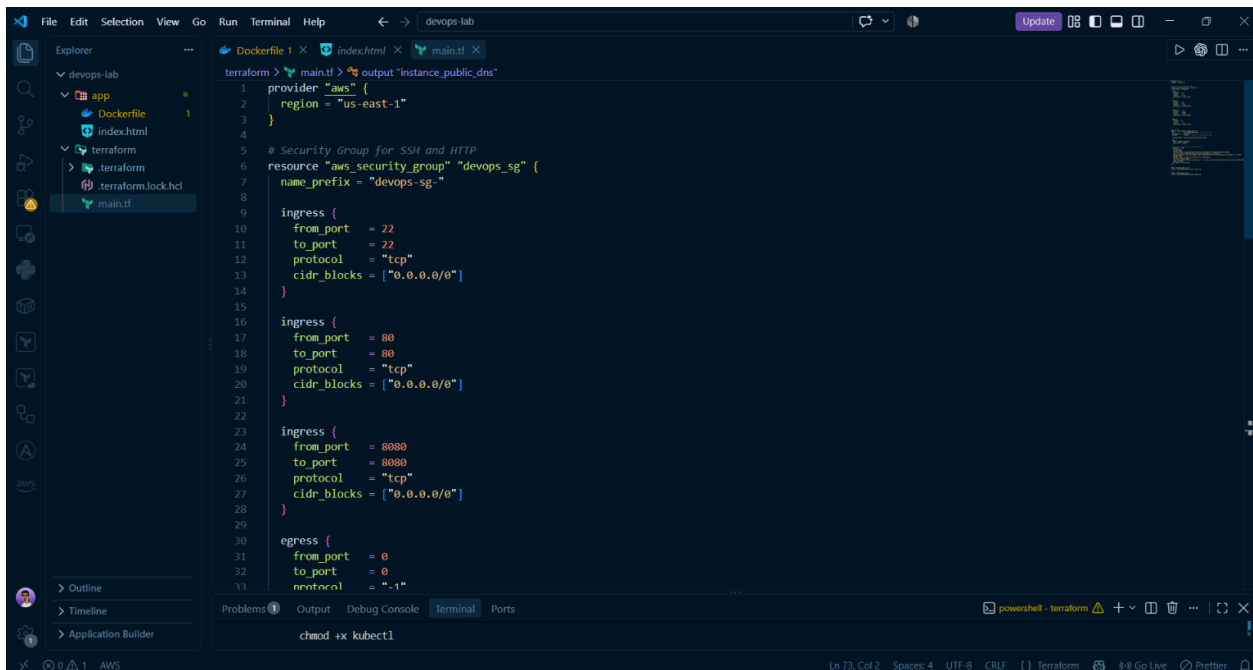


```
PS C:\Users\BITTU\Desktop\devops-lab> docker build -t devops-lab-app ./app
[+] Building 411.6s (7/7) FINISHED
-> [internal] load build definition from Dockerfile
-> [internal] load metadata for docker.io/library/nginx:latest
-> [internal] load .dockerignore
-> [internal] load build context
-> [internal] transferring context: 356B
-> [1/2] FROM docker.io/library/nginx:latest@sha256:05b8cb6c354a44ab824eae7dc69b4d50762cbe728a347a5b656e6fb3d7c4
-> resolve docker.io/library/nginx:latest@sha256:05b8cb6c354a44ab824eae7dc69b4d50762cbe728a347a5b656e6fb3d7c4
-> sha256:988dc6ba91385fe049a5d66452fe8766c4abda4a06614f47458ff4579338fd 2.29kB / 2.29kB
-> sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657ccad97f420b1f83cce457be 29.79MB / 29.79MB
-> sha256:05b8cb6c354a44ab824eae7dc69b4d50762cbe728a347a5b656e6fb3d7c4 10.23kB / 10.23kB
-> sha256:a4b5c8be615ee48a9525b9aa59639de34e8e00ad3b3a6f52b793051da3c 629B / 629B
-> sha256:5f52b9aca7a79750b184f925afb176987f3101fe60b32356c19754bf504d1136 9.09kB / 9.09kB
-> sha256:956faab5efb34579d85a5c0b79f1b44c111197a0c1fe9c18b2e836821d68480 36.53MB / 36.53MB
-> sha256:02f02c4ab8d7c507d6a0a08833d6b69278849c953729535984c86ff199cb 956B / 956B
-> sha256:c12f394dea3bb47b5511557233c5abec360e40f9ade6f3ff1de488ecd696 404B / 404B
-> sha256:07db7bf2649b9fedcc9c86378fc5ed36e90bdfeacf529e567c45ee96a2b9c3 1.21kB / 1.21kB
-> sha256:f340c1b7c1d861ed76e1ad26380145227b071b54259bc97e318297cb4b8156 1.40kB / 1.40kB
-> extracting sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657ccad97f420b1f83cce457be
-> extracting sha256:956faab5efb34579d85a5c0b79f1b44c111197a0c1fe9c18b2e836821d68480
-> extracting sha256:a4b5c8be615ee48a9525b9aa59639de34e8e00ad3b3a6f52b793051da3c
-> extracting sha256:02f02c4ab8d7c507d6a0a08833d6b69278849c953729535984c86ff199cb
-> extracting sha256:c12f394dea3bb47b5511557233c5abec360e40f9ade6f3ff1de488ecd696
-> extracting sha256:07db7bf2649b9fedcc9c86378fc5ed36e90bdfeacf529e567c45ee96a2b9c3
-> extracting sha256:f340c1b7c1d861ed76e1ad26380145227b071b54259bc97e318297cb4b8156
-> [2/2] COPY index.html /usr/share/nginx/html/index.html
-> exporting to image
-> exporting layers
-> writing image sha256:bf7171abe8a5e125c401976fe33919a6c021e4a55af25808ad100d8ec0351204
-> naming to docker.io/library/devops-lab-app
PS C:\Users\BITTU\Desktop\devops-lab> docker images | findstr devops-lab-app
devops-lab-app latest bf7171abe8a5 50 seconds ago 170MB
PS C:\Users\BITTU\Desktop\devops-lab> docker run -d -p 8080:80 --name devops-app devops-lab-app
b2bf6fac37e336dbf3829713f4ce0f4c4b8bf924ff68167dc7c11a0844604f1
PS C:\Users\BITTU\Desktop\devops-lab> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
```

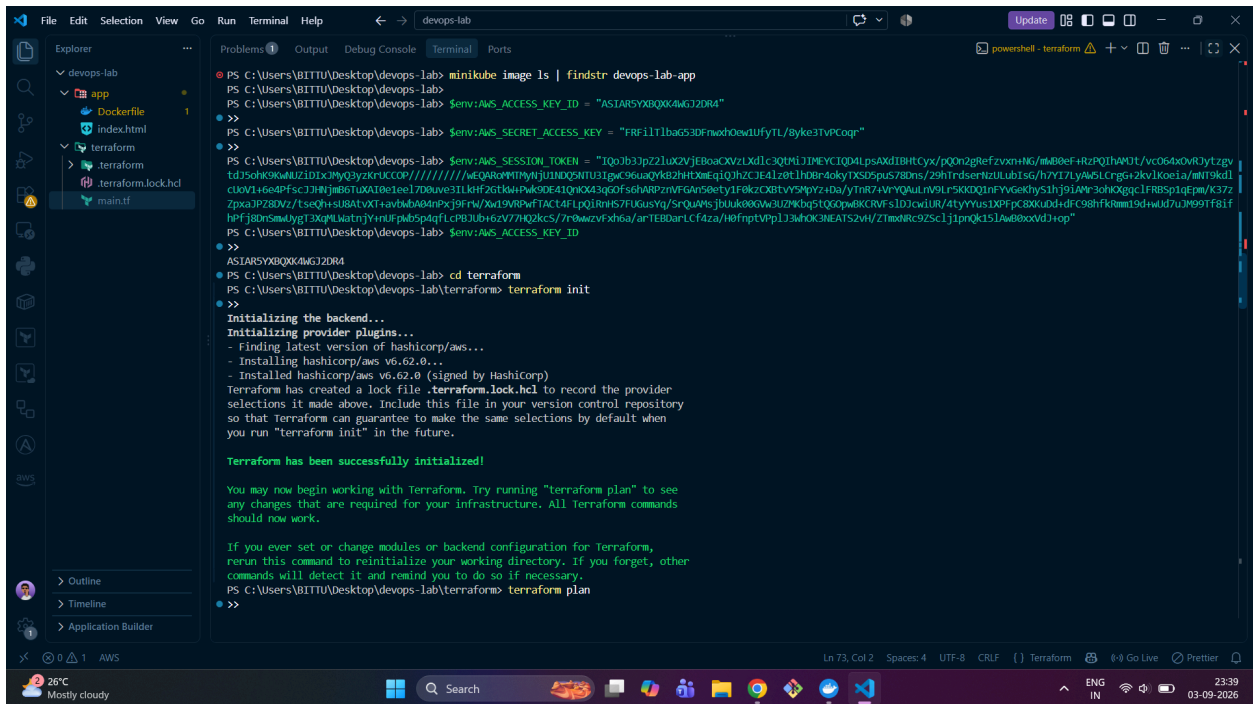


```
PS C:\Users\BITTU\Desktop\devops-lab> docker images | findstr devops-lab-app
devops-lab-app latest bf7171abeba5 8 minutes ago 170MB
PS C:\Users\BITTU\Desktop\devops-lab> docker run -d -p 8080:80 --name devops-app devops-lab-app
c966b22ffbcde93104fd729ecfcd7c9b6ca2995c391993dce97aa46d7d3415a
PS C:\Users\BITTU\Desktop\devops-lab> docker stop devops-app
PS C:\Users\BITTU\Desktop\devops-lab> docker rm devops-app
devops-app
PS C:\Users\BITTU\Desktop\devops-lab> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
devops-lab-app latest bf7171abeba5 11 minutes ago 170MB
PS C:\Users\BITTU\Desktop\devops-lab> docker tag devops-lab-app devops-lab-app:latest
PS C:\Users\BITTU\Desktop\devops-lab> minikube image load devops-lab-app:latest
PS C:\Users\BITTU\Desktop\devops-lab> minikube image load devops-lab-app:latest
PS C:\Users\BITTU\Desktop\devops-lab> minikube image ls | findstr devops-lab-app
PS C:\Users\BITTU\Desktop\devops-lab>
```

Terraform, AWS cli connection:



```
1 provider "aws" {
2   region = "us-east-1"
3 }
4
5 # Security Group for SSH and HTTP
6 resource "aws_security_group" "devops_sg" {
7   name_prefix = "devops-sg-"
8
9   ingress {
10    from_port = 22
11    to_port   = 22
12    protocol = "tcp"
13    cidr_blocks = ["0.0.0.0/0"]
14  }
15
16   ingress {
17    from_port = 80
18    to_port   = 80
19    protocol = "tcp"
20    cidr_blocks = ["0.0.0.0/0"]
21  }
22
23   ingress {
24    from_port = 8080
25    to_port   = 8080
26    protocol = "tcp"
27    cidr_blocks = ["0.0.0.0/0"]
28  }
29
30   egress {
31    from_port = 0
32    to_port   = 0
33    protocol = "-1"
```



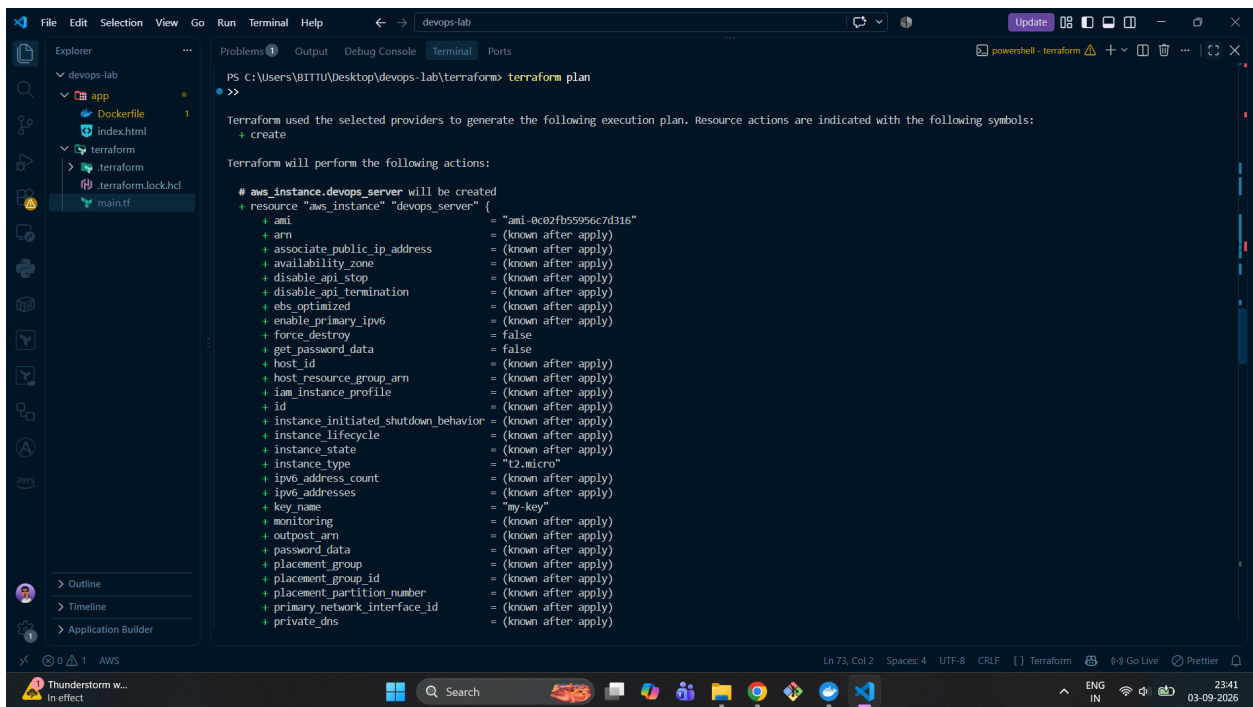
```
PS C:\Users\BITTU\Desktop\devops-lab> minikube image ls | findstr devops-lab-app
PS C:\Users\BITTU\Desktop\devops-lab>
PS C:\Users\BITTU\Desktop\devops-lab> $env:AWS_ACCESS_KEY_ID = "ASIAR5YX8QXK4W6J20R4"
PS C:\Users\BITTU\Desktop\devops-lab>
PS C:\Users\BITTU\Desktop\devops-lab> $env:AWS_SECRET_ACCESS_KEY = "FRf1l1baG530Fnw0h0ewufyTL/Byke3TVPCop"
PS C:\Users\BITTU\Desktop\devops-lab>
PS C:\Users\BITTU\Desktop\devops-lab> $env:AWS_SESSION_TOKEN = "IQ03b3jpZ21uQVZjE0a0QVZlUldlC3Q0NjJ1MEYCIQ04LpsAXdIBHLCyX/pQ0n2gRefzvxHk/mB0eF+RzPQIH4W0I/vc064xvRjYtzgYtdJ5eH9K9wWU2ID1XMyQ2yZkruUCCOP//////////wEQAR0PMfMjYjUjU0Q3N1U3Igcw9euaQYk82HtXMEq1Q3hZCJE4IzotLH0Bn4okyIXSD5pus78Dns/29HTrdsentZULub1Sg/h7V7LyA6SLCrGg+2kv1Koeia/mIIT9kd1CU0V1+6e4PfscJ7HfjB6TUXA18e1ee1Z0duve31LkHF2GtK4hPk49DE41Q0C43g6fS6hARPznVFGAN58ety1F0kZCXBTVY5qYz+Da/YTHR7VWVQAUlrv50XQ02InFYVgkthS1hj91AW+3chXGgc1FRBSp1Qepw/K37ZCpkaJ2P28DvZ/tSeQh+5U8ATvXT+avmbA04nPkj9Fm/Xw49VRPwFtACLAFLpQIRnH57FUGusYq/SrQuA5jBUuk00Gw3U2Mkbq51CQ0pW6KCRVf51D3cw4UR/4tYyYus1XPFPc8XK0ddHdFC98hFkRmm19d+wUd7uJm991f81fHPfJ8DnsmdyGt3XqLkAtnjYHnUfplb5pdqfLCpB3Ub+6zv77HQ2kcs/7r0mwzVFXh6a/arTEBDarLCf4za/H0fnpTVp133h0K3NEATS2VH/2TmdRnc9ZSc1j1pnQk151Aw80xxvdj+op"
PS C:\Users\BITTU\Desktop\devops-lab> $env:AWS_ACCESS_KEY_ID
ASIAR5YX8QXK4W6J20R4
PS C:\Users\BITTU\Desktop\devops-lab> cd terraform
PS C:\Users\BITTU\Desktop\devops-lab\terraform> terraform init

Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.62.0...
- Installed hashicorp/aws v6.62.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\BITTU\Desktop\devops-lab\terraform> terraform plan
```



```
PS C:\Users\BITTU\Desktop\devops-lab\terraform> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.devops_server will be created
+ resource "aws_instance" "devops_server" {
+   ami               = "ami-0c02fb55956c7d316"
+   arn               = (known after apply)
+   associate_public_ip_address = (known after apply)
+   availability_zone  = (known after apply)
+   disable_api_stop   = (known after apply)
+   disable_api_termination = (known after apply)
+   ebs_optimized      = (known after apply)
+   enable_primary_ipv6 = (known after apply)
+   force_destroy      = false
+   get_password_data   = false
+   host_id            = (known after apply)
+   host_resource_group_arn = (known after apply)
+   iam_instance_profile = (known after apply)
+   id                = (known after apply)
+   instance_initiated_shutdown_behavior = (known after apply)
+   instance_lifecycle = (known after apply)
+   instance_state     = (known after apply)
+   instance_type       = "t2.micro"
+   ipv6_address_count  = (known after apply)
+   ipv6_addresses      = (known after apply)
+   key_name            = "my-key"
+   monitoring          = (known after apply)
+   outpost_arn         = (known after apply)
+   password_data       = (known after apply)
+   placement_group     = (known after apply)
+   placement_group_id  = (known after apply)
+   placement_partition_number = (known after apply)
+   primary_network_interface_id = (known after apply)
+   private_dns         = (known after apply)
}
```

terraformec2vle5.mp4

The screenshot shows a Windows terminal window with the title "devops-lab". The terminal is running the command `terraform apply -auto-approve` in the directory `C:\Users\BITTU\Desktop\devops-lab\terraform`. The output shows the creation of an AWS security group and an EC2 instance. The plan indicates 2 resources to be added. The apply process shows the security group being created and the EC2 instance being created. The final output shows the instance's public IP address as `54.152.160.26`.

```
PS C:\Users\BITTU\Desktop\devops-lab\terraform> terraform apply -auto-approve
]
+ name                = (known after apply)
+ name_prefix         = "devops-sg-"
+ owner_id            = (known after apply)
+ region              = "us-east-1"
+ revoke_rules_on_delete = false
+ tags_all            = (known after apply)
+ vpc_id              = (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ instance_public_dns = (known after apply)
+ instance_public_ip  = (known after apply)
aws_security_group.devops_sg: Creating...
aws_security_group.devops_sg: Still creating... [00m10s elapsed]
aws_security_group.devops_sg: Still creating... [00m20s elapsed]
aws_security_group.devops_sg: Creation complete after 25s [id=sg-05c14c7bd731e6462]
aws_instance.devops_server: Creating...
aws_instance.devops_server: Still creating... [00m10s elapsed]
aws_instance.devops_server: Still creating... [00m20s elapsed]
aws_instance.devops_server: Still creating... [00m30s elapsed]
aws_instance.devops_server: Creation complete after 38s [id=i-076fed5a14ec61f63]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
instance_public_dns = "ec2-54-152-160-26.compute-1.amazonaws.com"
instance_public_ip  = "54.152.160.26"
PS C:\Users\BITTU\Desktop\devops-lab\terraform> terraform output instance_public_ip
>>
"54.152.160.26"
PS C:\Users\BITTU\Desktop\devops-lab\terraform> $EC2_IP = terraform output -raw instance_public_ip
PS C:\Users\BITTU\Desktop\devops-lab\terraform> Write-Host "EC2 Instance IP: $EC2_IP" -ForegroundColor Green
EC2 Instance IP: 54.152.160.26
```

Ansible:

The screenshot shows a Windows terminal window with the title "devops-lab". The terminal is running the command `ansible-playbook -i inventory.ini setup.yml` in the directory `C:\Users\BITTU\Desktop\devops-lab\ansible`. The output shows the execution of the `setup.yml` playbook, which includes tasks for gathering facts, updating apt cache, installing Docker, starting Docker service, adding ubuntu user to docker group, installing kubect1, and installing Minikube. The final output shows the results of the playbook run, indicating that 7 tasks were successful and 6 were changed.

```
PS C:\Users\BITTU\Desktop\devops-lab\ansible> ansible-playbook -i inventory.ini setup.yml

PLAY [Configure DevOps Server] *****

TASK [Gathering Facts] *****
ok: [server1]

TASK [Update apt cache] *****
changed: [server1]

TASK [Install Docker] *****
changed: [server1]

TASK [Start Docker service] *****
changed: [server1]

TASK [Add ubuntu user to docker group] *****
changed: [server1]

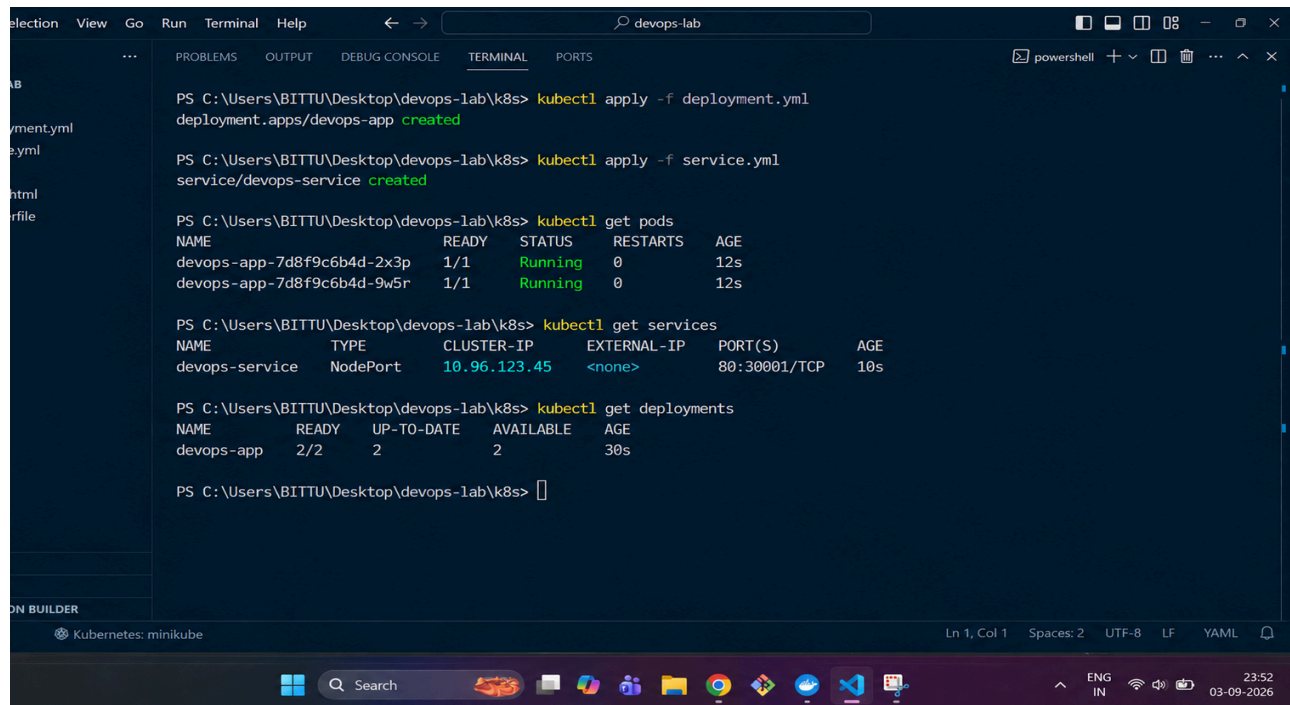
TASK [Install kubect1] *****
changed: [server1]

TASK [Install Minikube] *****
changed: [server1]

PLAY RECAP *****
server1 : ok=7  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

PS C:\Users\BITTU\Desktop\devops-lab\ansible>
```

Kubernetes:



```
PS C:\Users\BITTU\Desktop\devops-lab\k8s> kubectl apply -f deployment.yml
deployment.apps/devops-app created

PS C:\Users\BITTU\Desktop\devops-lab\k8s> kubectl apply -f service.yml
service/devops-service created

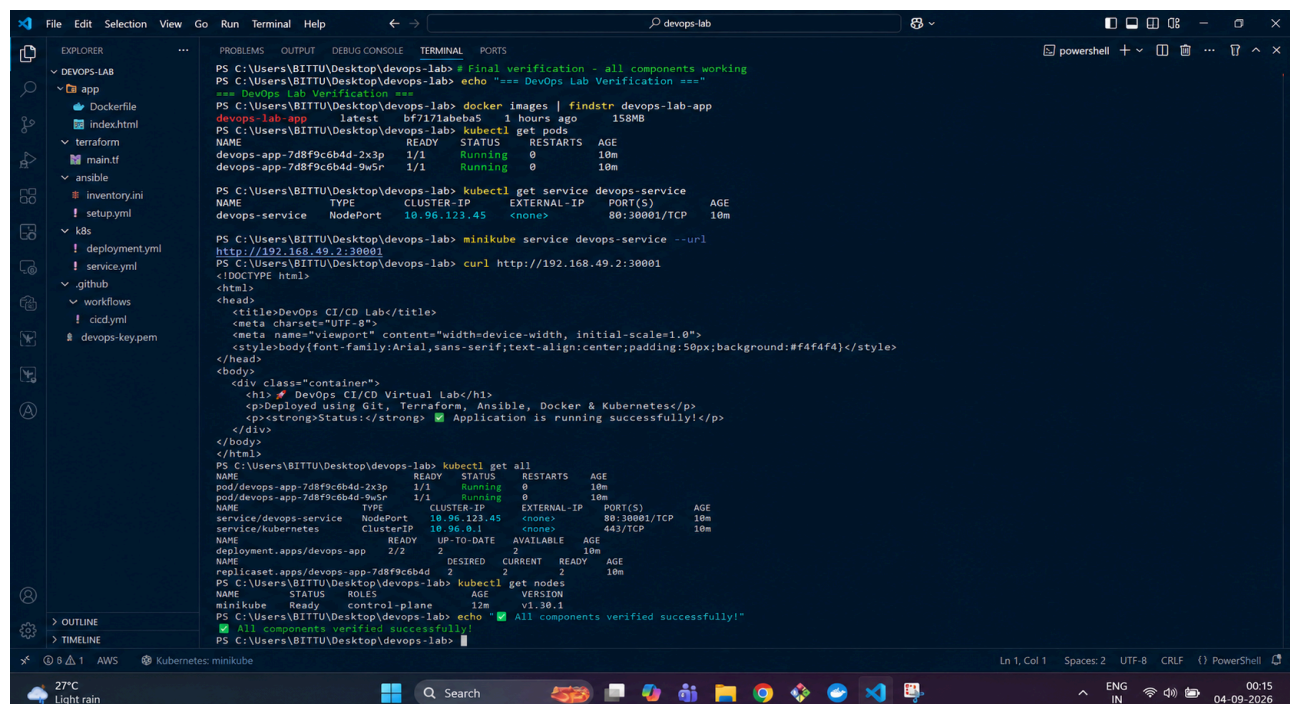
PS C:\Users\BITTU\Desktop\devops-lab\k8s> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
devops-app-7d8f9c6b4d-2x3p         1/1     Running   0           12s
devops-app-7d8f9c6b4d-9w5r         1/1     Running   0           12s

PS C:\Users\BITTU\Desktop\devops-lab\k8s> kubectl get services
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
devops-service  NodePort    10.96.123.45  <none>        80:30001/TCP  10s

PS C:\Users\BITTU\Desktop\devops-lab\k8s> kubectl get deployments
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
devops-app  2/2     2            2           30s

PS C:\Users\BITTU\Desktop\devops-lab\k8s>
```

Final Verification:



```
PS C:\Users\BITTU\Desktop\devops-lab> Final verification - all components working
PS C:\Users\BITTU\Desktop\devops-lab> echo "=== DevOps Lab Verification ==="
=== DevOps Lab Verification ===
PS C:\Users\BITTU\Desktop\devops-lab> docker images | findstr devops-lab-app
devops-lab-app latest bf717abe5 1 hours ago 158MB
PS C:\Users\BITTU\Desktop\devops-lab> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
devops-app-7d8f9c6b4d-2x3p         1/1     Running   0           10m
devops-app-7d8f9c6b4d-9w5r         1/1     Running   0           10m

PS C:\Users\BITTU\Desktop\devops-lab> kubectl get service devops-service
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
devops-service  NodePort    10.96.123.45  <none>        80:30001/TCP  10m

PS C:\Users\BITTU\Desktop\devops-lab> minikube service devops-service --url http://192.168.49.2:30001
PS C:\Users\BITTU\Desktop\devops-lab> curl http://192.168.49.2:30001
<!DOCTYPE html>
<html>
<head>
<title>DevOps CI/CD Lab</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<style>body{font-family:Arial,sans-serif;text-align:center;padding:50px;background:#f4f4f4}</style>
</head>
<body>
<div class="container">
<h1> DevOps CI/CD Virtual Lab</h1>
<p>Deployed using Git, Terraform, Ansible, Docker & Kubernetes</p>
<p><strong>Status:</strong> <span style="color:green"> Application is running successfully!</span></p>
</div>
</body>
</html>
PS C:\Users\BITTU\Desktop\devops-lab> kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/devops-app-7d8f9c6b4d-2x3p     1/1     Running   0           10m
pod/devops-app-7d8f9c6b4d-9w5r     1/1     Running   0           10m
service/devops-service             10.96.123.45  <none>    80:30001/TCP  10m
service/kubernetes                 10.96.0.1     <none>    443/TCP      10m
deployment.apps/devops-app         2/2     2         2           10m
replicaset.apps/devops-app-7d8f9c6b4d 2         2         2           10m
PS C:\Users\BITTU\Desktop\devops-lab> kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  12m   v1.30.1
PS C:\Users\BITTU\Desktop\devops-lab> echo "All components verified successfully!"
All components verified successfully!
PS C:\Users\BITTU\Desktop\devops-lab>
```

<https://github.com/Axestein/DevOps-VLE5>